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Unsteady Darcy Brinkman flow over a semi-infinite flat plate in the presence of thermo diffusion, radiation, and heat source/sink: RSM analysis

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ABSTRACT

This study investigates an unsteady convective flow of a ternary nanofluid (MgO–TiO2–CoFe2O4/H2O) over an infinite vertical plate through Darcy-Brinkman porous medium under the influences of a uniform transverse magnetic field, thermal radiation, Soret effect with different heat source/sinks. The novel contribution of this study lies in examining the effects of suspending ternary nanoparticles on heat and mass transfer in fluid flow. The governing coupled non-linear partial differential equations of the flow model are transformed into ordinary differential equations and solved using perturbation technique. Results reveal that ternary nanoparticles enhance the skin friction coefficient from 0.38% to 0.54%, while the Nusselt number decreases from 0.18% to 0.73%, and the Sherwood number increases significantly from 0.53% to 1.94%. The Soret effect significantly enhances species diffusion, with higher Schmidt numbers reducing molecular diffusion rates. RSM-based predictive models demonstrate excellent accuracy with R^2 values of 97.49% for skin friction, 99.94% for heat transfer, and 96.57% for mass transfer, validated through ANOVA.

1. Introduction

Conventional heat transfer fluids like water, ethylene glycol, and mineral oil have relatively low thermal performance compared to most solid materials. Fluids are often used as a heat carrier in heat transfer equipment. Nanofluids, a cutting-edge class of engineered colloids, comprise a base fluid infused with nanoparticles (1-100 nm), typically metals or metal oxides, dramatically enhancing thermal conductivity and convective heat transfer, highlighting nanoscale dimensions through comparative analysis with millimeter and micrometer scales. Accurate thermal conductivity measurement requires homogeneous, stable suspension and equilibrium conditions. These fluids are widely used in various engineering applications, including cooling systems, biomedical treatments, and energy harvesting, and have demonstrated remarkable thermal and electrical properties, making them suitable for complex heat transfer applications. In convective heat transfer studies, the Darcy-Brinkman model plays a crucial role in understanding the behaviour of nanofluids in porous media. This model incorporates both Darcy resistance and viscous effects, offering a more comprehensive approach to analysing fluid flow and thermal interactions. Ternary nanofluids, formed by suspending diverse nanoparticles in a base fluid, display novel thermophysical attributes that arise from the complex interplay of their constituent components. Investigating such systems enables researchers to explore new possibilities for optimizing heat transfer in industrial and technological applications.

Mythreye et al. (Mythreye et al., 2015) studied unsteady magneto-hydrodynamic (MHD) convective heat and mass transfer, which has been conducted by various researchers. The effects of chemical reaction on convective flows have been investigated. However, the combined influence of heat absorption, permeability, and moving plate on unsteady MHD convective flows with chemical reaction remains relatively unexplored. The study by Samad and Mansur-Rahman (Samad et al., 2006) investigates the impact of thermal radiation on unsteady MHD flow over a vertical porous plate embedded in a porous medium and analyzes heat and mass transfer characteristics under various physical parameters, revealing that increasing thermal radiation enhances heat transfer, while being subjected to a magnetic field suppresses fluid velocity. Singh et al. (Singh et al., 2017) explore the effects of Hall and ion-slip currents on unsteady MHD free convective flow of a rotating fluid over an oscillating vertical plate, and analyze the combined effects

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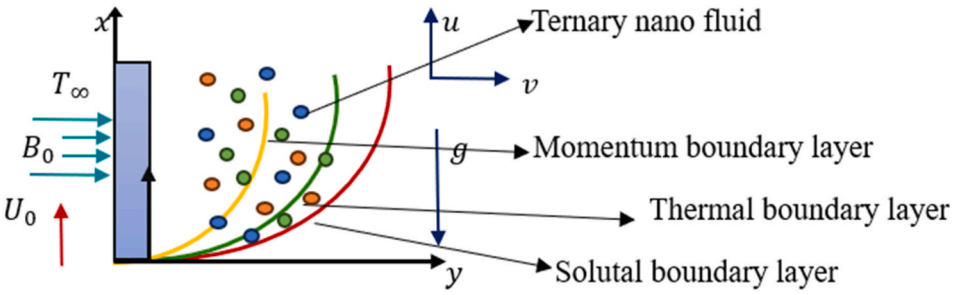


Fig. 1. Schematic sketch of the physical configuration.

of magnetic fields, rotation, and oscillatory motion on velocity and temperature distributions. Their findings indicate that Hall and ion-slip parameters significantly alter the fluid dynamics, impacting the heat transfer rate and flow stability. Seddeek and Salama (Seddeek et al., 2007) analysed that variable suction and temperature-dependent properties significantly impact the flow and heat transfer characteristics. The results showed that temperature-dependent viscosity and thermal conductivity enhance heat transfer rates. Kim et al. (Kim, 2000) analysed the impact of suction velocity, magnetic field, and plate velocity on flow and heat transfer. The results revealed that variable suction significantly influences the flow and heat transfer characteristics. Mabood et al. (Mabood et al., 2015) investigate that increasing the magnetic field strength suppresses fluid velocity, while enhancing thermal conductivity improves heat transfer. Gurivireddy et al. (Gurivireddy et al., 2016) analysed the significant influence of thermal diffusion on the flow field and heat transfer rates. This research has implications for various engineering applications, including chemical reactors and heat exchangers. Ravikumar et al. (Ravikumar et al., 2014) investigated the simultaneous effect of heat absorption and velocity, and temperature profiles were significantly affected by the magnetic field and heat absorption. Provided insights into the behavior of non-Newtonian fluids subjected to magnetic fields and heat transfer. Goud et al. (Goud et al., 2019) examine thermal radiation significantly enhances heat transfer, while the porous medium increases flow resistance. This research contributes to the understanding of MHD flow in engineering applications, with subsequent studies incorporating nanofluids, chemical reactions, and variable viscosity for more advanced models. Er-Raki et al. (Er-Raki et al., 2006) explore the influence of the Soret effect on boundary layer flow within a porous enclosure with imposed horizontal heat and mass fluxes and analyze how thermal and solutal gradients affect flow behavior and mass transfer using numerical simulations. Their results reveal that the Soret effect has a profound impact on concentration profile and flow stability. Ahmed et al. (Ahmed, 2010) investigated the influences of Soret and Dufour effects on flow and heat transfer characteristics. The results also showed that the magnetic field and porosity impact the flow behavior. Wang and Tan (Wang et al., 2011) investigate the stability of double-diffusive convection induced by Soret effects in a Maxwell fluid within a porous structure and reveal that the Soret effect significantly influences the onset of convection, while fluid elasticity alters the stability characteristics. Ibrahim et al. (Ibrahim et al., 2008) analysed that the chemical reaction and radiation absorption significantly altered the flow patterns and heat transfer mechanisms. Singh and Singh (Singh et al., 2012) examined how the magnetic field exerted a profound influence on the flow behavior. Dharmaiah et al. (Dharmaiah et al., 2018) analysed the radiation absorption and heat source significantly influenced the flow and heat transfer characteristics. The diffusion thermo effect was essential to the flow behavior. Reddy et al. (Reddy, 2017) investigated significant influences of chemical reactions and radiation absorption substantially impact flow and heat transfer behavior, while viscous dissipation significantly boosts heat transfer rates. Venkateswaralu and Satyanarayana (Venkateswarlu et al., 2015) investigated the effects of chemical reaction and radiation absorption on nanofluid flow, heat, and mass transfer within a rotating framework. Jakeer et al. (Jakeer et al., 2023) explored the influence of induced magnetic fields and double-diffusive convection on Carreau nanofluid flows in various geometries, utilizing a comparative numerical and artificial neural network analysis. Bilal et al. (Bilal et al., 2022) numerically investigated the dynamics of thermal enhancement mechanisms in ternary MHD hybrid nanofluid flows over various surfaces, such as plane sheets, cones, and wedges. They found that the ternary hybrid nanofluid exhibited significant thermal enhancement compared to conventional fluids. The addition of multiple nanoparticles in a base fluid was found to enhance the thermal conductivity and viscosity of the fluid, leading to improved heat transfer performance. Jakati et al. (Jakati et al.) studied MHD boundary layer flow of nanofluid over a moving surface in the presence of thermal radiation, employing the Homotopy Analysis Method showed that nanoparticles enhance thermal conductivity, while magnetic fields slow down the flow. Thermal radiation significantly impacts temperature profiles. The study provides insights into flow and heat transfer characteristics. Vajravelu et al. (Vajravelu et al., 2014) analysed MHD flow and heat transfer of an Ostwald-de Waele fluid over an unsteady stretching surface, found that magnetic field and unsteadiness significantly impact flow and heat transfer showed that velocity and temperature increase with Ostwald-de Waele parameter. Mishra et al. (Mishra et al., 2021) explore that heat generation enhances thermal boundary layer thickness, while increasing the magnetic field suppresses fluid velocity. Abel and Mahesha (Abel et al., 2008) investigate that increasing the magnetic field strength suppresses the fluid velocity, while variable thermal conductivity and heat sources significantly impact thermal boundary layer thickness. Seshaiah et al. (Seshaiah et al., 2013) examine the effects of chemical reactions and radiation on unsteady MHD free convective fluid flow within a porous medium, accounting for time-dependent suction and a temperature gradient heat source, and analyze the way of these factors influence velocity, temperature, and concentration profiles, revealing that increasing radiation enhances heat transfer, while chemical reactions significantly affect concentration distributions. Dharmaiah et al. (Dharmaiah et al., 2020) investigate how thermophoretic forces enhance mass transfer, while Hall currents significantly impact the flow behavior (Farooq et al., 2025a, 2025b; Noreen et al., 2025b, 2025c). aim to study the develop the combined influence of thermal radiation, nanoparticle transport, bioconvection due to microorganisms, and chemical reaction activation energy on heat and mass transfer characteristics.

In light of the aforementioned discussions, the present study aims to investigate the Soret, radiation, and magnetic field effects on convective Darcy-Brinkman ternary nanofluid flow over an infinite plate. The analysis will account for the presence of different heat sources and sinks, providing a comprehensive understanding of the underlying transport phenomena. Results of this study contribute to the development of more efficient thermal management systems.

1.1. Research motivation

Despite considerable progress in the study of MHD flows, heat and mass transfer, and nanofluid dynamics in porous media, the comprehensive understanding of ternary nanofluid systems under complex physical effects remains limited. Most existing studies have focused on mono or hybrid nanofluids, often neglecting the synergistic thermal advantages offered by ternary nanoparticle suspensions. Furthermore, the simultaneous consideration of heat generation and absorption mechanisms such as constant, temperature-dependent, and temperature-gradient-dependent sources/sinks alongside thermo-diffusion (Soret effect), thermal radiation, and electromagnetic forces within the Darcy-Brinkman porous framework is rarely addressed. The interplay between these phenomena significantly alters the transport behavior, yet remains underexplored in current literature. Moreover, while numerical and analytical techniques are commonly employed, the application of RSM for statistically modeling and optimizing such multifactorial systems has received insufficient attention. This creates a notable gap in developing predictive, validated models for advanced thermal management and heat transfer enhancement using ternary nanofluids in porous, radiative, and magnetized environments.

1.2. Application

In present study explored the multifaceted applications of nanofluids incorporating Magnesium Oxide (MgO), Titanium Oxide (TiO_2), and Carbonate Ferrite ($CoFe_2O_4$), each offering distinct advantages across thermal, environmental, and biomedical domains. These nanoparticles, when dispersed in base fluids, significantly enhance heat transfer capabilities, making them ideal for cooling systems in electronics, automotive engines, and industrial heat exchangers. MgO and TiO_2 nanofluids, with their superior thermal conductivity, are particularly effective in solar thermal systems, improving heat absorption and boosting the efficiency of solar collectors. TiO_2 also exhibits strong photocatalytic activity under UV light, enabling its use in environmental cleanup applications such as wastewater treatment, where it degrades harmful pollutants. Meanwhile, $CoFe_2O_4$ nanofluids, due to their magnetic properties, are employed in MHD power generation systems, where magnetic fields control fluid flow to optimize energy conversion. These magnetic characteristics also make cobalt ferrite nanoparticles suitable for targeted drug delivery, allowing precise control in therapeutic applications. In enhanced oil recovery (EOR), MgO and TiO_2 nanofluids improve fluid mobility by reducing viscosity and increasing thermal conductivity, facilitating more efficient extraction processes. Research on ternary nanofluids ($MgO–TiO_2–CoFe_2O_4$) offers practical applications in microelectronics and renewable energy. In microelectronics, their enhanced thermal conductivity and magnetic field control via $CoFe_2O_4$ enable advanced cooling for high-power chips and GPUs. In renewable energy, superior heat absorption and photocatalytic properties boost solar thermal collectors and water purification, while MHD traits aid power generation and geothermal extraction. The study delivers an optimized framework for efficient thermal systems in electronics, solar, and heat exchangers.

1.3. Objective of the work

This research is aimed at providing comprehensive solutions to address the following questions.

- The influence of ternary nanoparticles on flow, heat and mass transfer in presence of magnetic field with porous medium.
- The impacts of thermal radiation, temperature dependent and temperature gradient-dependent heat source/sink in presence of nanoparticle suspension on the temperature field.
- How do concentration profiles vary due to the variations of first order chemical reaction, Schmidt number, and Soret effect?

By addressing these objectives, this research can gain valuable insights into the performance, stability, and potential applications of multi-hybrid nanofluids in various heat transfer systems.

2. Formulation of the problem

A theoretical analysis is conducted to investigate the unsteady natural convection flow of a ternary nanofluid over a vertical, permeable, semi-infinite moving plate, incorporating the effects of an internal heat source. As illustrated in Fig. 1, the flow is oriented along the x-axis, parallel to the plate, with the y-axis perpendicular to it. An external magnetic field of strength B_0 is applied transversely to the flow direction, generating Lorentz forces that modulate fluid motion. Under the low magnetic Reynolds number approximation, the induced magnetic field and external electric field due to charge polarization are assumed to be negligible. Initially, at $t = 0$, the plate and the nanofluid are in thermal and solutal equilibrium at the temperature T_0 and concentration C_0. For $t > 0$ The plate temperature and species concentration oscillate sinusoidally about a steady-state mean.

The working medium is a water-based ternary nanofluid consisting of MgO, TiO_2, and $CoFe_2O_4$ nanoparticles, which are assumed to be monodispersed and spherically symmetric. The nanoparticle-fluid suspension is modeled under the continuum hypothesis, assuming a homogeneous mixture where Brownian motion and thermophoresis effects are negligible. The fluid phase and nanoparticles are considered to be in local thermal equilibrium, and the effective thermophysical properties of the nanofluid are evaluated using established mixing models. The governing equations incorporate the effects of porous drag, MHD interactions, thermal radiation, and the Soret effect, which induces species diffusion in response to temperature gradients. The Darcy-Brinkman model is employed to account for momentum transport through the porous medium, incorporating viscous and permeability effects. Due to the semi-infinite nature of the plate, all field variables are assumed to be functions of the transverse coordinate y and time t only. The problem formulation includes continuity, momentum, energy, and species transport equations, subject to appropriate boundary conditions at the plate and at infinity. The dimensionless form of these equations is solved numerically to analyze the influence of key governing parameters on velocity, thermal distribution, and species concentration.

Assumptions of the model:

- The plate is long enough in x-direction, so that all the physical quantities are independent of x except pressure.
- The flow is driven by thermal buoyancy forces.
- Induced magnetic field and applied electrical field are neglected.
- Viscous, Darcy, and Homic Dissipation effects are ignored.

Utilizing the boundary layer approximations, the nanofluid flow is mathematically described by the following governing equations (Prasad et al. (Prasad et al., 2018), Seshaiah et al. (Seshaiah et al., 2013))

$$\frac{\partial v'}{\partial y'} = 0 \tag{1}$$

$$\rho_{tnf}\left(\frac{\partial u'}{\partial t'} + v'\frac{\partial u'}{\partial y'}\right) = \mu_{tnf}\frac{\partial^2 u'}{\partial y'^2} + (\rho\beta)_{tnf} g\left(T' - T'_\infty\right) - \frac{\mu_{tnf}}{K'}u' - \sigma_{tnf}B_0^2 u' \tag{2}$$

$$\left(\rho C_p\right)_{tnf}\left(\frac{\partial T'}{\partial t'} + v'\frac{\partial T'}{\partial y'}\right) = K_{tnf}\frac{\partial^2 T'}{\partial y'^2} - \frac{\partial q_r}{\partial y'} + a_1 q_1 + a_2 q_2\left(T' - T'_\infty\right) + a_3 q_3\frac{\partial\left(T' - T'_\infty\right)}{\partial y'} \tag{3}$$

$$\frac{\partial C'}{\partial t'} + v'\frac{\partial C'}{\partial y'} = D_B\frac{\partial^2 C'}{\partial y'^2} - K_1\left(C' - C'_\infty\right) + \frac{D_m K_T}{T_m}\frac{\partial^2 T'}{\partial y'^2} \tag{4}$$

The velocity components along the x and y axes are denoted by u' and

Table 1
Thermophysical properties of ternary nanofluid (Madkhali et al., 2023).

Density	$\rho_{tnf} = (1-\phi_3)\left[(1-\phi_2)\left((1-\phi_1)\rho_{bf} + \phi_1\rho_1\right) + \phi_2\rho_2\right] + \phi_3\rho_3$
Thermal conductivity	$K_{nf} = K_f\left[\frac{K_3 + 2K_{bf} - 2\phi_3(K_{bf} - K_3)}{K_3 + 2K_{bf} + \phi_3(K_{bf} - K_3)}\right]$
	$K_{hnf} = K_{nf}\left[\frac{K_2 + 2K_{nf} - 2\phi_2(K_{nf} - K_2)}{K_2 + 2K_{nf} + \phi_2(K_{nf} - K_2)}\right]$
	$K_{tnf} = K_{hnf}\left[\frac{K_1 + 2K_{hnf} - 2\phi_1(K_{hnf} - K_2)}{K_1 + 2K_{hnf} + \phi_1(K_{hnf} - K_2)}\right]$
Viscosity	$\mu_{tnf} = \frac{\mu_{bf}}{(1-\phi_1)^{2.5}(1-\phi_2)^{2.5}(1-\phi_3)^{2.5}}$
Electrical conductivity	$\sigma_{nf} = \sigma_{bf}\left[\frac{\sigma_3(1+2\phi_3) + 2\sigma_{bf}(1-\phi_3)}{\sigma_3(1-\phi_3) + \sigma_{bf}(2+\phi_3)}\right]$
	$\sigma_{hnf} = \sigma_{nf}\left[\frac{\sigma_2(1+2\phi_2) + 2\sigma_{nf}(1-\phi_2)}{\sigma_2(1-\phi_2) + \sigma_{nf}(2+\phi_2)}\right]$
	$\sigma_{tnf} = \sigma_{hnf}\left[\frac{\sigma_1(1+2\phi_1) + 2\sigma_{hnf}(1-\phi_1)}{\sigma_1(1-\phi_1) + \sigma_{hnf}(2+\phi_1)}\right]$
Heat Capacitance	$(\rho C_p)_{tnf} = (1-\phi_1)\left[(1-\phi_2)\left((1-\phi_3)(\rho C_p)_{bf} + \phi_3(\rho C_p)_3\right) + \phi_2(\rho C_p)_2\right] + \phi_1(\rho C_p)_1$
Thermal Expansion	$(\rho\beta)_{tnf} = (1-\phi_1)\left[(1-\phi_2)\left((1-\phi_3)(\rho\beta)_{bf} + \phi_3(\rho\beta)_3\right) + \phi_2(\rho\beta)_2\right] + \phi_1(\rho\beta)_1$
Solutal Expansion	$(\rho\beta')_{tnf} = (1-\phi_1)\left[(1-\phi_2)\left((1-\phi_3)(\rho\beta')_{bf} + \phi_3(\rho\beta')_3\right) + \phi_2(\rho\beta')_2\right] + \phi_1(\rho\beta')_1$

Table 2
Thermophysical values of TiO_2, $CoFe_2O_4$, MgO nanofluids with base fluid (Bilal et al., 2022).

Base fluid and nano particles	ρ (kg/m^3)	k(W/mK)	Cp(j/kgK)	σ (S/m)	β (1/k)
Distilled water (H_2O)	997.1	0.613	4179	0.05	21×10^5
Magnesium Oxide (MgO)	4907	3.7	700	5.51×10^9	1.26×10^5
Titanium Oxide (TiO_2)	4250	8.9538	686.2	2.38×10^6	0.9×10^5
Cobalt Ferrite ($CoFe_2O_4$)	3560	45	955	1.42×10^{-3}	3.9×10^6

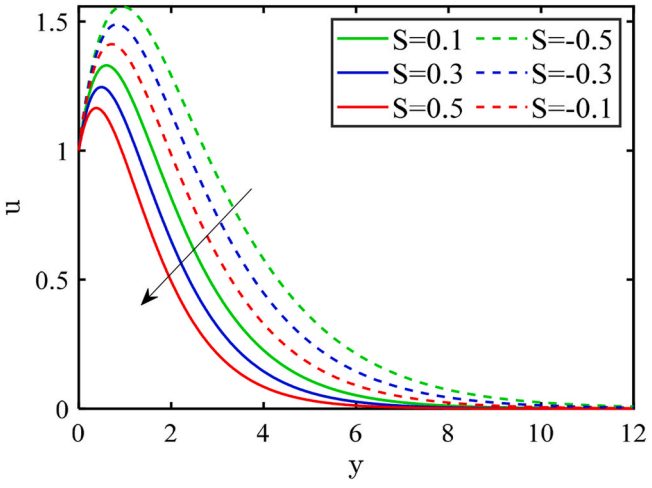


Fig. 2. Velocity profile for S

ν' respectively. The nanofluid's thermophysical properties include β_{tnf}, the coefficient of thermal expansion; σ_{tnf}, the electrical conductivity; ρ_{tnf}, the density; μ_{tnf}, the viscosity; $(\rho C_p)_{tnf}$, the heat capacitance; α_{tnf}, the thermal diffusivity; and K_{tnf}, the thermal conductivity. α_{tnf} The thermal diffusivity is given as $\alpha_{thf} = \frac{K_{tnf}}{(\rho C_p)_{tnf}}$. In the energy equation (3), the third,

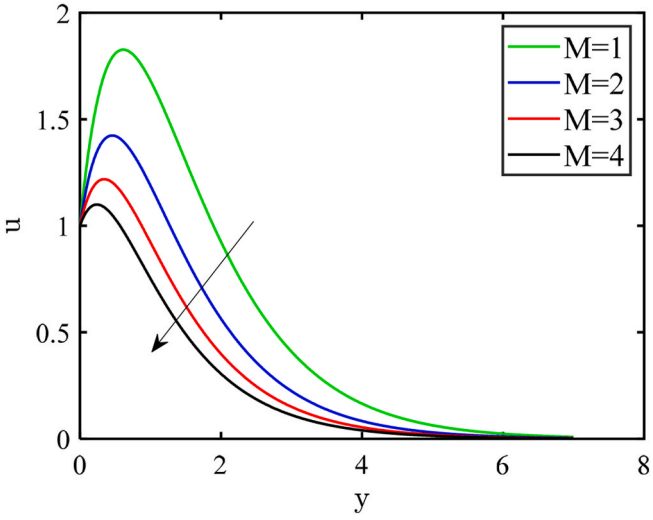


Fig. 3. Velocity profile for M

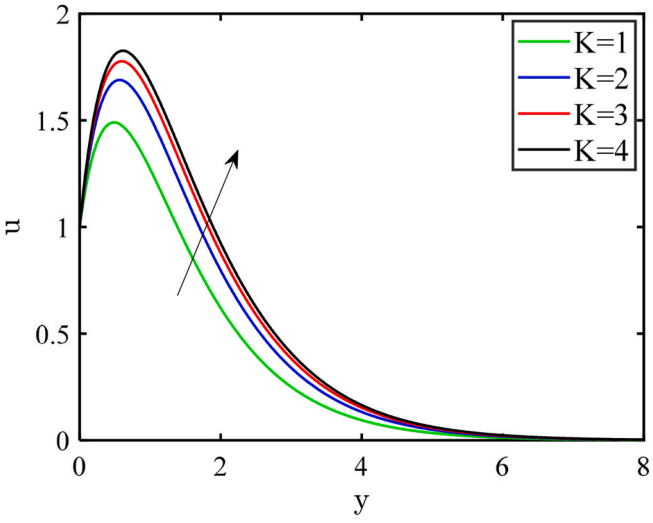


Fig. 4. Velocity profile for K

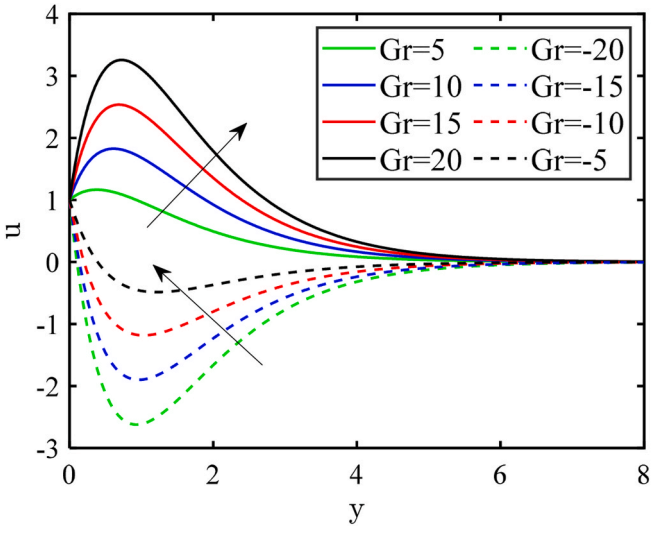


Fig. 5. Velocity profile for G_r

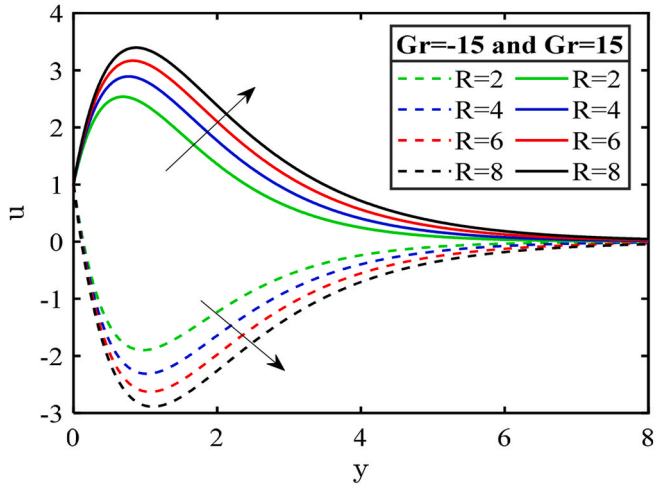


Fig. 6. Velocity profile for R

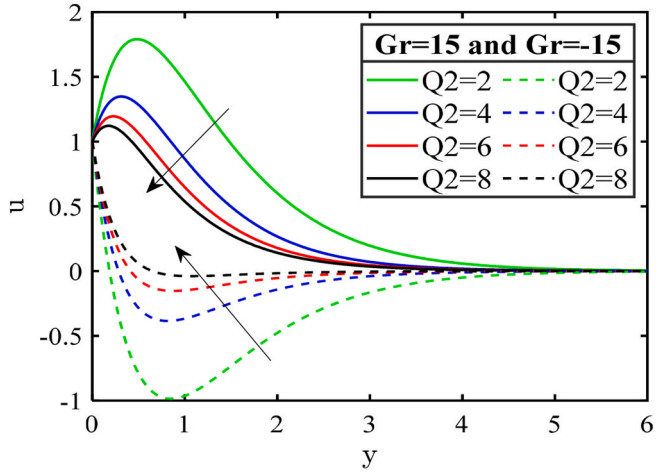


Fig. 7. Velocity profile for Q_2

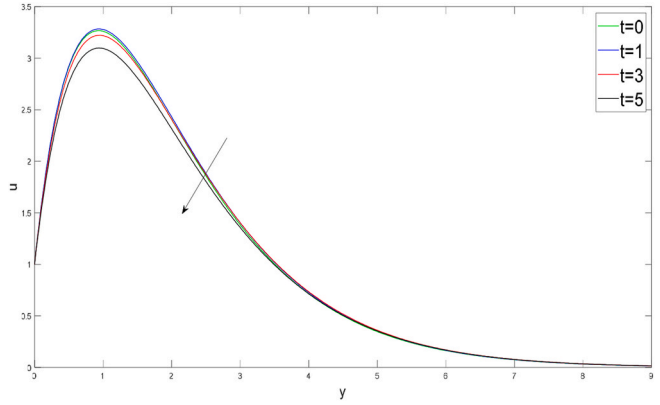


Fig. 8. Velocity profile for t

fourth and fifth term of RHS represents q_1 is constant heat source ($a_1 = 1, a_2 = 0, a_3 = 0$), q_2 is temperature-dependent heat source ($a_1 = 0, a_2 = 1, a_3 = 0$), and q_3 is the temperature gradient heat source ($a_1 = 0, a_2 = 0, a_3 = 1$). Additionally, ϕ represents the solid volume fraction of nanoparticles. The physical properties of the base fluid (distilled water) and the nanoparticles (MgO, TiO_2, and $CoFe_2O_4$) used for validation purposes are summarized in Table 1 and Table 2. The

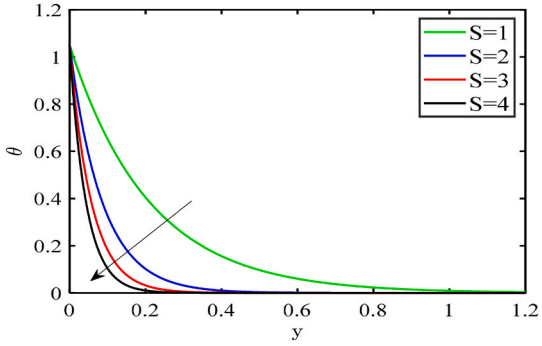


Fig. 9. Temperature profile for S

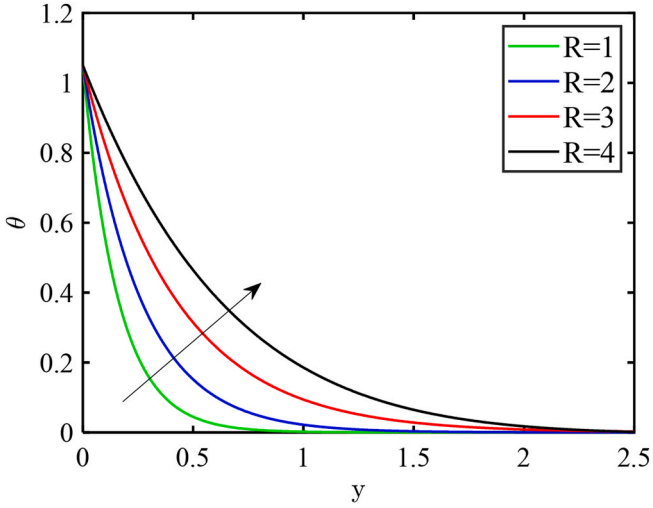


Fig. 10. Temperature profile for R

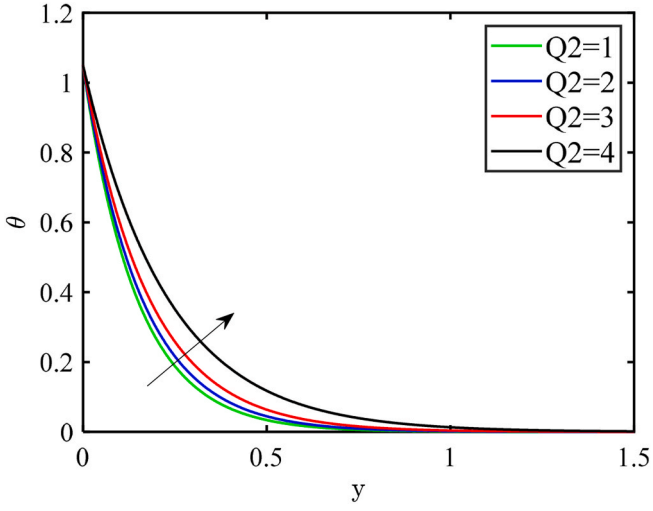


Fig. 11. Temperature profile for Q_2

boundary conditions governing the problem are as follows:

$$\begin{aligned} & t' < 0, u'(y', t') = 0, T' = T'_\infty, C' = C'_\infty \\ & t' \geq 0, u'(y', t') = U_0, T' = T'_w + (T'_w - T'_\infty)\varepsilon e^{iw't'}, \\ & C' = C'_w + (C'_w - C'_\infty)\varepsilon e^{iw't'} \text{ at y' } = 0 \\ & u'(y', t') = 0, T' = T'_\infty, C' = C'_\infty \text{ as } y' \to \infty \end{aligned} \quad (5)$$

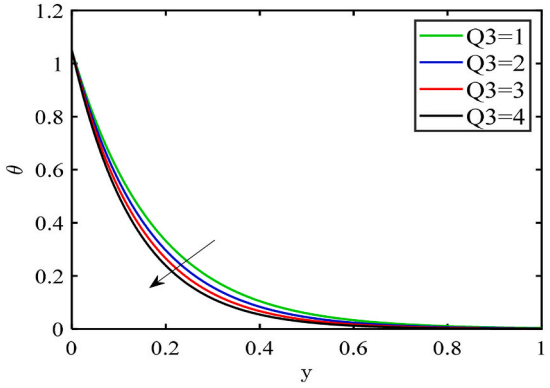


Fig. 12. Temperature profile for Q_3

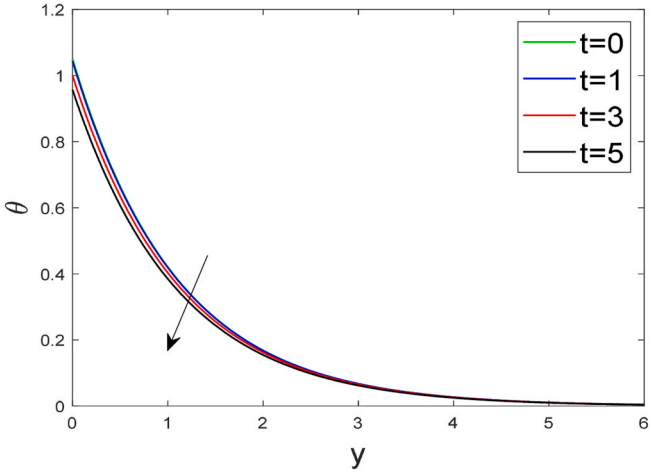


Fig. 13. Temperature profile for t

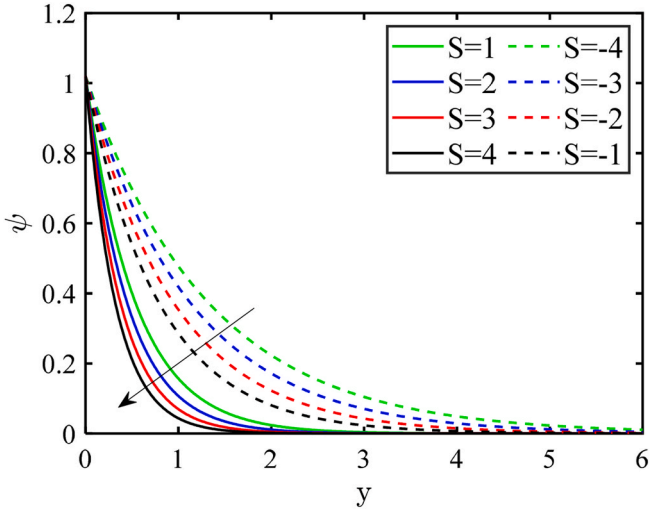


Fig. 14. Concentration profile for S

In this study, the boundary conditions are defined to simulate oscillatory thermal and solutal effects at the wall. Initially, for ($t' < 0$), the fluid remains stationary with uniform ambient temperature and concentration. At ($t' \geq 0$), the wall velocity is set to (U_0), while temperature and concentration fluctuate sinusoidally around their respective wall values. Far from the wall ($y' \to \infty$) The fluid returns to its

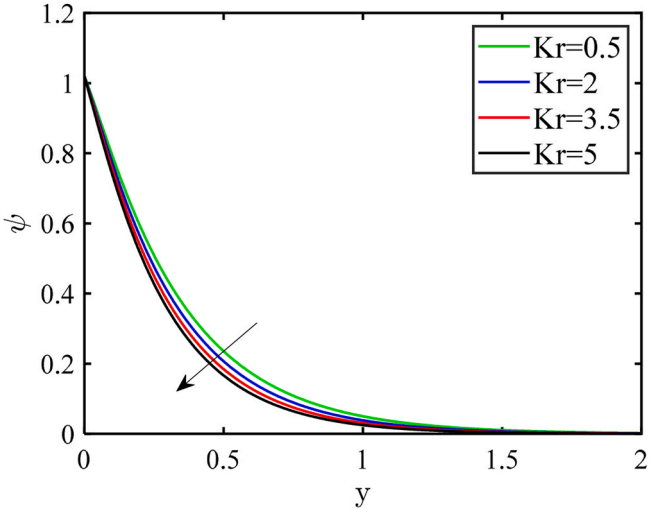


Fig. 15. Concentration profile for K_r

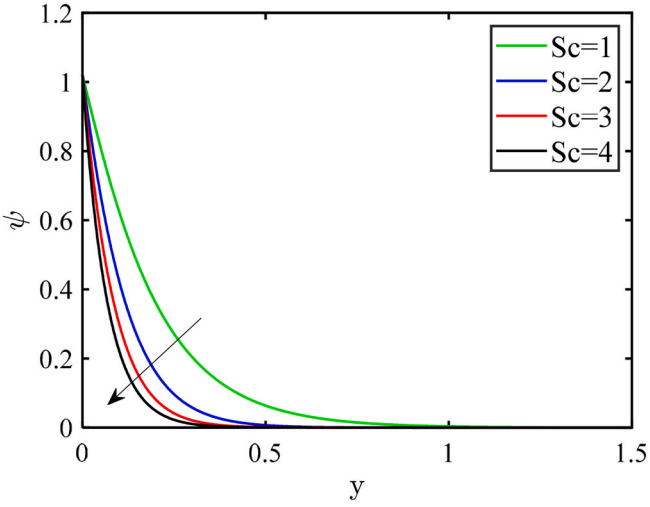


Fig. 16. Concentration profile for S_c

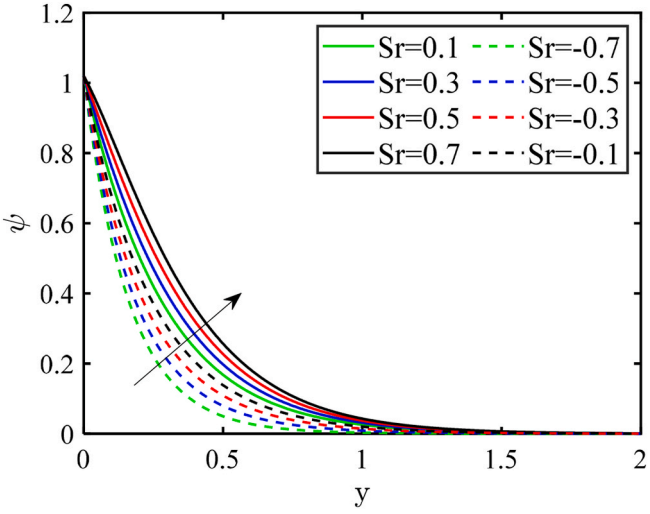


Fig. 17. Concentration profile for S_r

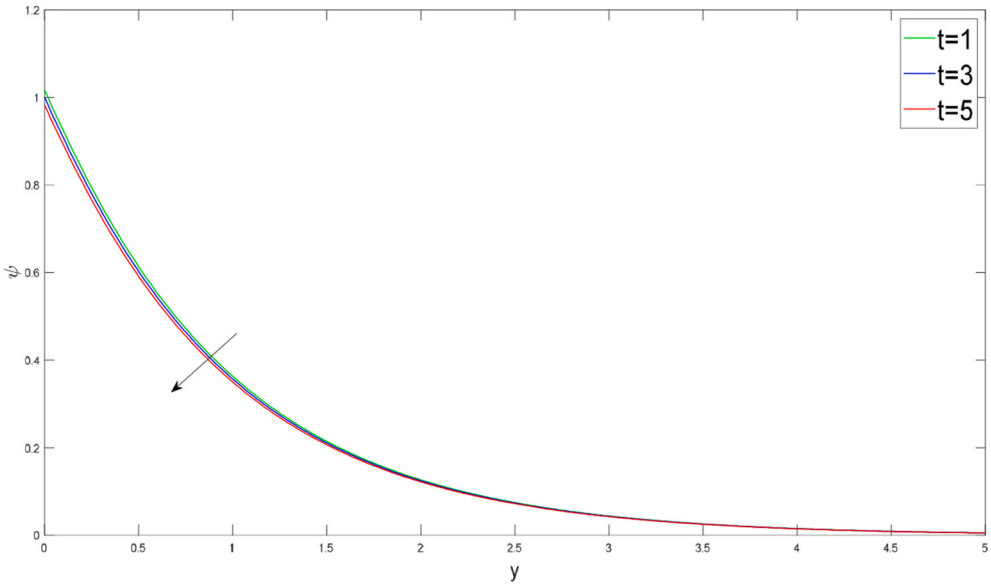


Fig. 18. Concentration profile for t.

Table 3
Numerical values of skin-friction coefficient (cf) with $S_c = 0.60$; $w = 0.1$; $P_r = 7$; $K_r = 0.5$

G_r	t	M	K	S	R	Q_2	Q_3	$-f''(0)$		
								MgO/H_2O	TiO_2/H_2O	$CoFe_2O_4/H_2O$
1	0	1	4	0.1	1	2	2	0.65201	0.67671	0.67809
2								1.46790	1.51870	1.52030
3								1.13430	1.18960	1.19200
4								0.80066	0.86061	0.86360
	1							0.65202	0.67672	0.67809
	2							0.65212	0.67682	0.67800
	3							0.65232	0.67702	0.67840
		2						1.10450	1.13910	1.14030
		3						1.44770	1.48960	1.49060
		4						1.73360	1.78140	1.78240
			5					0.62321	0.64853	0.64992
			6					0.60359	0.62935	0.63075
			7					0.58935	0.61544	0.61685
				0.3				0.75872	0.78384	0.78593
				0.5				0.87991	0.90529	0.90825
				0.7				1.01560	1.04110	1.04500
					2			0.64023	0.66519	0.66656
					3			0.62992	0.65510	0.65647
					4			0.62075	0.64613	0.64749
						3		0.73490	0.75805	0.75936
						4		0.79015	0.81243	0.81368
						5		0.82991	0.85163	0.85284
							3	0.57107	0.59757	0.59897
							4	0.51356	0.54147	0.54290
							5	0.47137	0.50037	0.50184

ambient state, ensuring a stable far-field environment. Where T' is the local temperature.

At the plate, the velocity component $v' = -V_0$, V_0 is a constant representing the normal velocity; $V_0 > 0$ signifies suction, $V_0 < 0$ signifies blowing injection.

Let us define the following dimensionless variables:

$$u = \frac{u'}{U_0}, y = \frac{U_0 y'}{\nu_f}, t = \frac{U_0^2 t'}{\nu_f}, w = \frac{\nu_f w'}{U_0^2}, \theta = \frac{(T' - T'_\infty)}{(T'_w - T'_\infty)}, S = \frac{V_0}{U_0}$$

$$K_r = \frac{K_f \nu_f}{U_0^2},\ S_c = \frac{\nu_f}{D_B},\ P_r = \frac{\nu_f}{\alpha_f}, K' = \frac{K \rho_f U_0^2}{\nu_f^2}, M = \frac{\sigma B_0^2 \nu_f}{\rho_f U_0^2} \tag{6}$$

$$\psi = \frac{(C' - C'_\infty)}{(C'_w - C'_\infty)}, S_r = \frac{D_m K_T (T'_w - T'_\infty)}{\nu T_m (C'_w - C'_\infty)}, G_r = \frac{(\rho\beta)_f g \nu_f (T'_w - T'_\infty)}{\rho_f U_0^3}$$

Here, $P_r = \frac{\nu_f}{\alpha_f}$ is the Prandtl number, S is the suction (S > 0) or injection (S < 0) parameter, M is the Magnetic parameter, K_r is the chemical reaction parameter, S_c is the Schmidt number, G_r is the Grashof number, K is the permeability parameter.

In the governing energy equation (3), the term $\frac{\partial q_r}{\partial y'}$ represents the radiative heat flux contribution. To model this effect accurately, the Rosseland approximation is often employed, which assumes that the medium is optically thick. Under this approximation, the radiative heat flux q_r is expressed as:

$$q_r = -\frac{4\sigma^*}{3k_\rho}\frac{\partial T^4}{\partial y'} \tag{7}$$

Table 4
Numerical values of Nusselt number (Nu) with $S_c = 0.60$; $w = 0.1$.

S	t	Q_2	P_r	Q_3	R	$-\theta'(0)$		
						MgO/H_2O	TiO_2/H_2O	$CoFe_2O_4/H_2O$
0.1	0	1	7	2	1	1.1755	1.1751	1.1757
0.3						1.2916	1.2923	1.2933
0.5						1.4340	1.4330	1.4345
0.7						1.6043	1.6041	1.6064
	1					1.1725	1.1723	1.1729
	2					1.1670	1.1675	1.1682
	3					1.1614	1.1609	1.1616
		2				2.3138	2.2137	2.2157
		3				3.2482	3.1547	3.1583
		4				4.1100	4.0213	4.0268
			6.2			1.6658	1.1654	1.1661
			0.71			0.8324	0.8302	0.8319
			8			1.1858	1.1848	1.1854
				3		0.8011	0.8007	0.8009
				4		0.6042	0.6040	0.6041
				5		0.4841	0.4839	0.4840
					3	2.2169	2.2160	2.2171
					5	3.2517	3.1603	3.1617
					7	4.1159	4.0308	4.0323

Where Stefan-Boltzmann constant is represented by σ^* and k_ρ is the mean absorption coefficient. For the small temperature difference within the flow are sufficiently small, by Taylor series expansion can be employed to T^4 approximate, resulting in a linearized expression of the form (Pantokratoras et al., 2014)

$$T^4 \approx 4T_\infty^3 - 3T_\infty^4 \tag{8}$$

By using (7) and (8)

$$\frac{\partial q_r}{\partial y} = -\frac{16\sigma^* T_\infty^3}{3k_\rho}\frac{\partial^2 T}{\partial y^2} \tag{10}$$

In view of equations (6) to (10), equations (2)–(4) reduce to the following dimensionless form:

$$L_3\left(\frac{\partial u}{\partial t} - S\frac{\partial u}{\partial y}\right) = L_4\frac{\partial^2 u}{\partial y^2} + L_5 G_r\theta - \left(\frac{L_4}{K} + L_6 M\right)u \tag{11}$$

$$L_1\left(\frac{\partial \theta}{\partial t} - S\frac{\partial \theta}{\partial y}\right) = \frac{1}{P_r}(L_2 + R)\frac{\partial^2 \theta}{\partial y^2} + \frac{1}{P_r}\left(a_1 Q_1 + a_2 Q_2\theta + a_3 Q_3\frac{\partial \theta}{\partial y}\right) \tag{12}$$

$$\frac{\partial \psi}{\partial t} - S\frac{\partial \psi}{\partial y} = \frac{1}{S_c}\frac{\partial^2 \psi}{\partial y^2} - K_r\psi + S_r\frac{\partial^2 \theta}{\partial y^2} \tag{13}$$

With boundary conditions

$$\begin{aligned} t < 0 &: u = 0, \theta = 0, \psi = 0 \\ t \geq 0 &: u = 1, \theta = 1 + \varepsilon e^{iwt} \ at\ y = 0 \\ &u = 0, \theta = 0, \psi = 0 \ as\ y \to \infty \end{aligned} \tag{14}$$

3. Solution of the problem

Various solutions techniques (Asghar et al., 2025; Noreen et al., 2025a; Shahzad et al., 2025) are used in fluid flow equations. For present study equations (11)–(13) constitute a set of coupled non-linear partial differential equations, whose closed-form solutions are challenging to derive. To facilitate the solution process, these equations are transformed into ordinary differential equations by superimposing the unsteady flow on the mean steady flow. Enabling the representation of velocity, temperature, and concentration in the neighborhood of the plate as;

$$\begin{aligned} u(y,t) &= u_0 + \varepsilon u_1 e^{iwt} \\ \theta(y,t) &= \theta_0 + \varepsilon u_1 e^{iwt} \\ \psi(y,t) &= \psi_0 + \varepsilon \psi_1 e^{iwt} \end{aligned} \tag{15}$$

Where $\varepsilon(\ll 1)$ is a parameter.

Equations (11)–(13) are reduced to

$$L_4 u_0'' + SL_3 u_0' - \left(\frac{L_4}{K} + L_6 M\right)u_0 = -L_6 G_r\theta_0 \tag{16}$$

$$L_4 u_1'' + SL_3 u_1' - \left[L_3(iw) + \left(\frac{L_4}{K} + L_6 M\right)\right]u_1 = -L_5 G_r\theta_1 \tag{17}$$

$$(L_2 + R)\theta_0'' + (P_r L_1 S - a_3 Q_3)\theta_0' - a_2 Q_2\theta_0 = 0 \tag{18}$$

$$(L_2 + R)\theta_1'' + (SP_r L_1 - a_3 Q_3)\theta_1' - (a_2 Q_2 + P_r L_1(iw))\theta_1 = 0 \tag{19}$$

Table 5
Numerical values of Sherwood Number (Sh) with $= 0.1$; $P_r = 7$

S_r	t	K_r	S_c	S	R	Q_2	Q_3	$-\psi'(0)$		
								MgO/H_2O	TiO_2/H_2O	CO_2Fe_4/H_2O
0.2	0	4	0.60	1	0.5	2	2	2.0790	2.0816	2.0789
0.3								2.1511	2.1549	2.1509
0.4								2.2231	2.2283	2.2230
0.5								2.2952	2.3017	2.2950
	1							1.8285	1.8290	1.8285
	2							1.8280	1.8285	1.8280
	3							1.8271	1.8276	1.8271
		5						2.3260	2.3291	2.3259
		6						2.5505	2.5540	2.5503
		7						2.7576	2.7615	2.7574
			0.22					1.0595	1.0598	1.0594
			0.30					1.2719	1.2723	1.2719
			0.78					3.0953	3.1203	2.5999
				3				2.1133	2.1221	2.1130
				5				1.8364	1.8598	1.8358
				7				1.4458	1.4857	1.4447
					1.5			2.2414	2.2463	2.2413
					2.5			2.4750	2.6422	2.6263
					3.5			6.2303	6.7117	6.2150
						3		1.8470	1.8536	1.8469
						4		1.7509	1.7517	1.7509
						5		1.7042	1.7051	1.7042
							3	1.8890	1.8891	1.8890
							4	1.9087	1.9088	1.9087
							5	1.9182	1.9182	1.9182

Table 6
Numerical values of skin friction, Nusselt number, and Sherwood number with $G_r = 1, M = 4, K = 4, R = 3.5, S = 1, Q_2 = 5; Q_3 = 3, S_c = 0.60, S_r = 0.5$ and $K_r = 7$.

ϕ_1	ϕ_2	ϕ_3	$-f''(0)$	$-\theta'(0)$	$-\psi'(0)$
0	0	0	2.3084	3.9660	1.5454
0.01	0.01	0.01	2.3171	3.9371	1.5754
0.02	0.02	0.02	2.3165	3.9066	1.6040
0.03	0.03	0.03	2.3092	3.8752	1.6312
0.04	0.04	0.04	2.2946	3.8429	1.6572
0.01	0.01	0	2.2960	3.9449	1.5677
0.02	0.02	0	2.2803	3.9226	1.5892
0.03	0.03	0	2.2616	3.8999	1.6101
0.04	0.04	0	2.2402	3.8766	1.6302
0.01	0	0.01	2.3004	3.9512	1.5615
0.02	0	0.02	2.2888	3.9356	1.5771
0.03	0	0.03	2.2738	3.9199	1.5922
0.04	0	0.04	2.2556	3.9040	1.6069
0	0.01	0.01	2.3493	3.9450	1.5675
0	0.02	0.02	2.3849	3.9229	1.5888
0	0.03	0.03	2.4154	3.9003	1.6094
0	0.04	0.04	2.4412	3.8772	1.6293
0.01	0	0	2.2781	3.9589	1.5536
0.02	0	0	2.2478	3.9512	1.5616
0.03	0	0	2.2175	3.9435	1.5694
0.04	0	0	2.1871	3.9359	1.5771
0	0.01	0	2.3272	3.9528	1.5596
0	0.02	0	2.3445	3.9387	1.5737
0	0.03	0	2.3603	3.9243	1.5874
0	0.04	0	2.3748	3.9097	1.6009
0	0	0.01	2.3316	3.9590	1.5534
0	0	0.02	2.3533	3.9513	1.5613
0	0	0.03	2.3732	3.9436	1.5691
0	0	0.04	2.3916	3.9359	1.5768

$$\psi_0'' + SS_c\psi_0' - K_rS_c\psi_0 = -S_rS_c\theta_0'' \tag{20}$$

$$\psi_1'' + SS_c\psi_1' - (K_rS_c + S_c(iw))\psi_1 = -S_cS_r\theta_1'' \tag{21}$$

The boundary conditions (14) become

$$\begin{aligned} &u_0 = 1, u_1 = 0, \theta_0 = 1, \theta_1 = 1, \psi_0 = 1, \psi_1 = 1 \ at\ y = 0 \\ &u_0 = 0, u_1 = 0, \theta_0 = 0, \theta_1 = 0, \psi_0 = 0, \psi_1 = 0 \ as\ y \to \infty \end{aligned} \tag{22}$$

Using the boundary conditions (22), solve equations (16)–(21), we get

$$u(y,t) = (1-b_3)e^{-m_5y} + b_3e^{-m_1y} + \varepsilon(b_4(e^{-m_2y} - e^{-m_6y}))e^{iwt} \tag{23}$$

$$\theta(y,t) = e^{-m_1y} + \epsilon(e^{-m_2y})e^{iwt} \tag{24}$$

$$\psi(y,t) = (1-b_1)e^{-m_3y} + b_1e^{-m_1y} + \epsilon((1-b_2)e^{-m_4y} + b_2e^{-m_2y})e^{iwt} \tag{25}$$

Skin friction, Rate of Heat transfer, and Mass transfer:

The dimensional expression for the Skin friction/shearing stress at the plate is

$$Cf = -\left(\frac{\partial u}{\partial t}\right)_{y=0} = (1-b_3)m_5 + b_3m_1 + \varepsilon b_4(m_2 - m_6)e^{iwt} \tag{26}$$

Similarly, the Nusselt number/rate of heat transfer at the plate is given by

$$Nu = -(1+R)\left(\frac{\partial\theta}{\partial t}\right)_{y=0} = (1+R)\left(m_1 + \varepsilon\, m_2e^{iwt}\right) \tag{27}$$

The Sherwood number/rate of mass transfer at the plate is given by

$$Sh = -\left(\frac{\partial\psi}{\partial t}\right)_{y=0} = m_3 + b_1(m_1 - m_3) + \varepsilon\,(m_4 + b_2(m_2 - m_4))e^{iwt} \tag{28}$$

4. Result and discussion

The current research examines micropolar fluid boundary layer flow with ternary nanoparticles, incorporating mass transport and thermal conductivity. Solution plots for various parameters are illustrated through graphs. Results exhibit major features of flow, heat, and mass transfer characteristics in terms of using nanoparticles. The inclusion of nanoparticles influences temperature, velocity, and concentration distributions along with skin friction, heat transfer, and mass transfer coefficients. Suction preserves a constant distribution of MgO, TiO_2, and $CoFe_2O_4$ nanoparticles.

Range of parameters : $0 \le t \le 5, 1 \le Sc \le 4, 0.5 \le Kr \le 5, -4 \le S \le 4, 1 \le M \le 4,$

$$-20 \le Gr \le 20, 1 \le R \le 8, 1 \le Q2 \le 8, -0.7 \le Sr \le 0.7, 1 \le Q3 \le 4,$$

Figs. 2–18 illustrate that velocity, temperature, and concentration profiles are significantly influenced by key physical parameters governing fluid flow and heat and mass transfer. Fig. 2 demonstrates that suction reduces boundary layer thickness, stabilizing the flow, whereas injection enhances velocity and mass transport. Fig. 3 shows that the presence of a magnetic field introduces a Lorentz force, which resists fluid motion, leading to a reduction in velocity. Figs. 4 and 5 show that increased permeability facilitates smoother flow, while a higher Grashof number strengthens buoyancy-driven convection, accelerating fluid motion. Figs. 6 and 7 direct radiation alter thermal gradients, contributing to velocity reduction, while temperature-dependent heat sources reduce viscosity, enhancing velocity. Fig. 8 shows an initial rise in velocity profile followed by a slow decline, eventually stabilizing over time.

Temperature profiles exhibit notable variations with different influencing factors. Fig. 9 observes that suction leads to a thinner thermal boundary layer, lowering fluid temperature, while injection retains heat, increasing temperature levels. Figss. 10–12 direct radiation enhances thermal energy transfer, leading to temperature rise, whereas temperature-dependent and gradient heat sources intensify localized heating, promoting stronger convection and heat diffusion. In Fig. 13, the temperature profile converges towards zero as time progresses due to the effect of heat sources.

Similarly, concentration profiles are affected by fluid motion and thermal effects. Suction decreases mass diffusion, while injection enhances species transport within the boundary layer observed in Fig. 14. Fig. 15 illustrates that chemical reactions play a vital role in modifying concentration by either consuming or generating species, thereby affecting mass transfer rates. Fig. 16 exemplifies higher Schmidt number reduces molecular diffusion, restricting species transport. Fig. 17 indicates that the Soret effect facilitates concentration gradients due to temperature differences, promoting species diffusion. Fig. 18 shows that the concentration profile follows a similar trend due to diffusion effects over time.

These interdependent effects have significant implications in industrial and engineering applications, particularly in heat exchangers, chemical processing, and energy systems. The findings of the study have relevance for numerous engineering fields, such as heat exchangers, chemical reactors, and electronic devices. The findings can be utilized for optimization of the system design and operation in order to increase efficiency and performance.

From Table 3, detailed analysis shows that increasing the suction parameter enhances flow stability, as reflected in higher $-f''(0)$ values across different nanofluids. Additionally, parameters such as thermal radiation, heat sources, and magnetic field effects influence the boundary layer characteristics. From Table 4, based on the numerical data for $-\theta'(0)$, the study highlights the effects of various parameters on the thermal boundary layer in the presence of different nanofluids. As S increases, $-\theta'(0)$ also increases, indicating enhanced heat transfer rates due to stronger suction. Additionally, higher values of Q_2, Q_3 and R further intensify the thermal gradient near the surface. Among the nanofluids, $CoFe_2O_4/H_2O$ consistently shows slightly higher heat

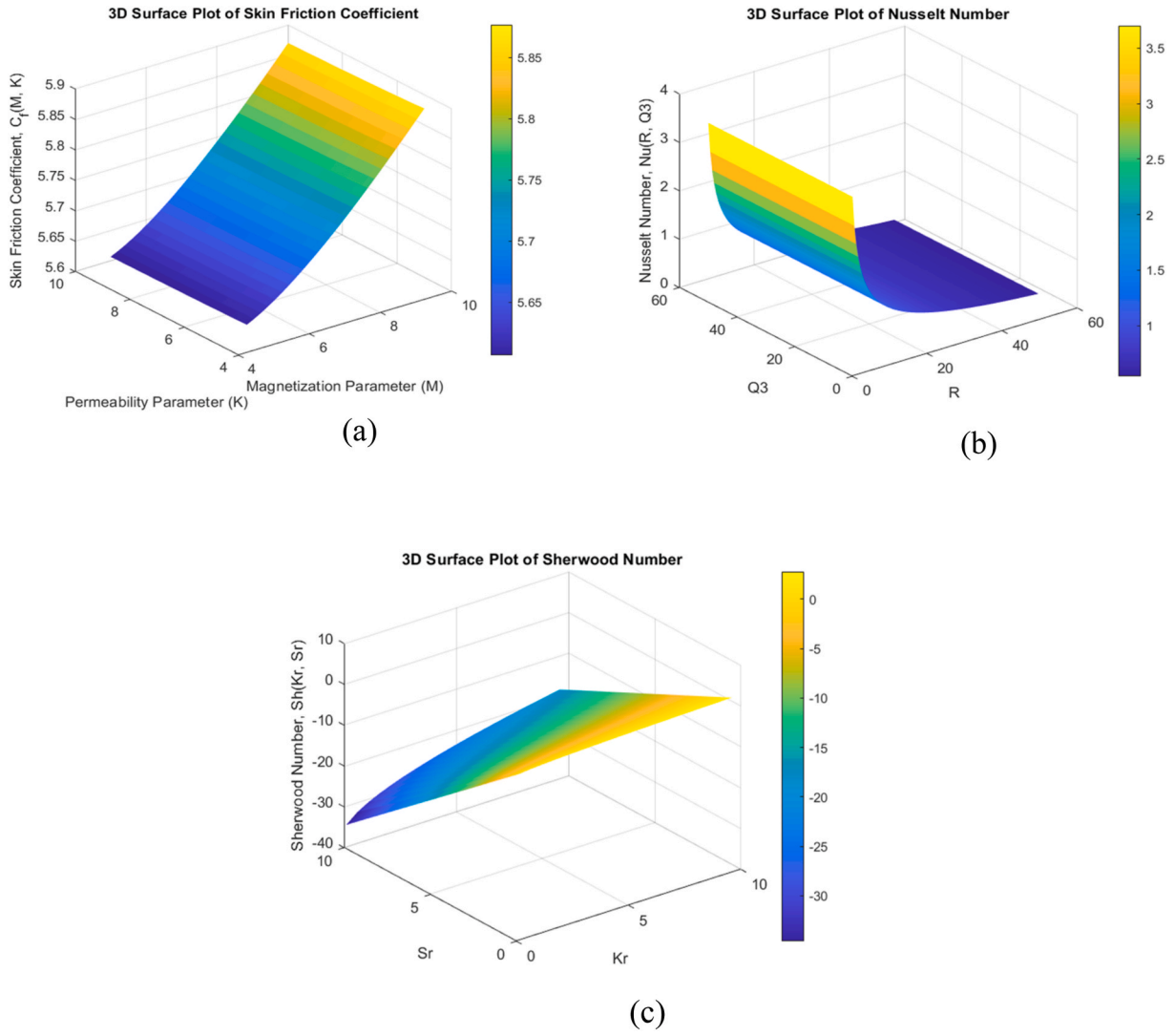


Fig. 19. 3-D graphs for skin friction, rate of heat and mass transfer.

transfer rates, making it thermally more effective under the given conditions. From Table 5, the analysis of $-\psi'(0)$ demonstrates that increasing suction and radiation parameters significantly enhance mass transfer rates, indicated by higher $-\psi'(0)$ values. Among the nanofluids, TiO_2/H_2O consistently exhibits slightly better mass transfer performance, though the difference is marginal. Overall, higher thermo diffusion and reaction rates further contribute to intensified nanoparticle transport near the surface. From Table 6, the updated observation in the presence and absence of nanoparticles shows that skin friction varies between 0.38% and 0.54%, indicating a minimal impact on fluid resistance. The Nusselt number decreases from 0.18% to 0.73%, confirming a slight reduction in heat transfer, while the Sherwood number increases from 0.53% to 1.94%, highlighting improved diffusion with ternary nanoparticles. In the presence of hybrid nanoparticles, skin friction varies from 0.54% to 0.18%, Nusselt number consistently drops between 0.53% and 0.73%, and Sherwood number improves from 1.45% to 1.94%. For single nanoparticles, MgO shows reduced skin friction due to enhanced cooling, whereas TiO_2 and $CoFe_2O_4$ exhibit slight increases. Overall, heat transfer decreases from 0.15% to 0.73%, while mass transfer rises from 0.53% to 1.94%, confirming the superior performance of ternary nanofluids. The velocity, temperature, and concentration profiles we used $w = 0.1; t = 0.01; P_r = 7; M = 0.5; \varepsilon = 0.02; S_c = 0.60; S = 0.1; K = 4; R = 5; K_r = 0.1; Q_1 = 0; Q_2 = 2; Q_3 = 0; a_1 = 0;\ a_2 = 1; a_3 = 0; \phi_1 = 0.01;\ \phi_2 = 0.02;$ and $\phi_3 = 0.03$.

4.1. Physical significance

The observed results provide the following insights:

Flow Control: Suction, injection, and magnetic fields can effectively regulate boundary layer thickness and velocity, enabling precise control in industrial and microfluidic applications. Interplay of suction, magnetic effects, heat sources, and nanoparticle types enables the optimization of skin friction, heat, and mass transfer to improve efficiency and performance.

Thermal Management: Ternary nanofluids allow tunable heat transfer, beneficial for electronics cooling, solar collectors, and chemical reactors.

Enhanced Mass Transport: Nanoparticles and thermo-diffusion mechanisms promote species transport, critical for catalytic processes and reactive flows.

Fig. 19(a) illustrates the skin friction coefficient $Cf(M, K)$ increases with both the M and K. The upward trend indicates that higher permeability and magnetic effects lead to an increase in frictional resistance within the boundary layer due to the Lorentz force generated by the applied magnetic field, which enhances resistance to fluid motion. The increase in skin friction also suggests that the fluid experiences greater shear stress as both parameters increase, making these effects crucial for flow stability analysis. Fig. 19(b) shows the 3D plot of the Nusselt number $Nu(R, Q_3)$ shows the rate of heat transfer. A significant

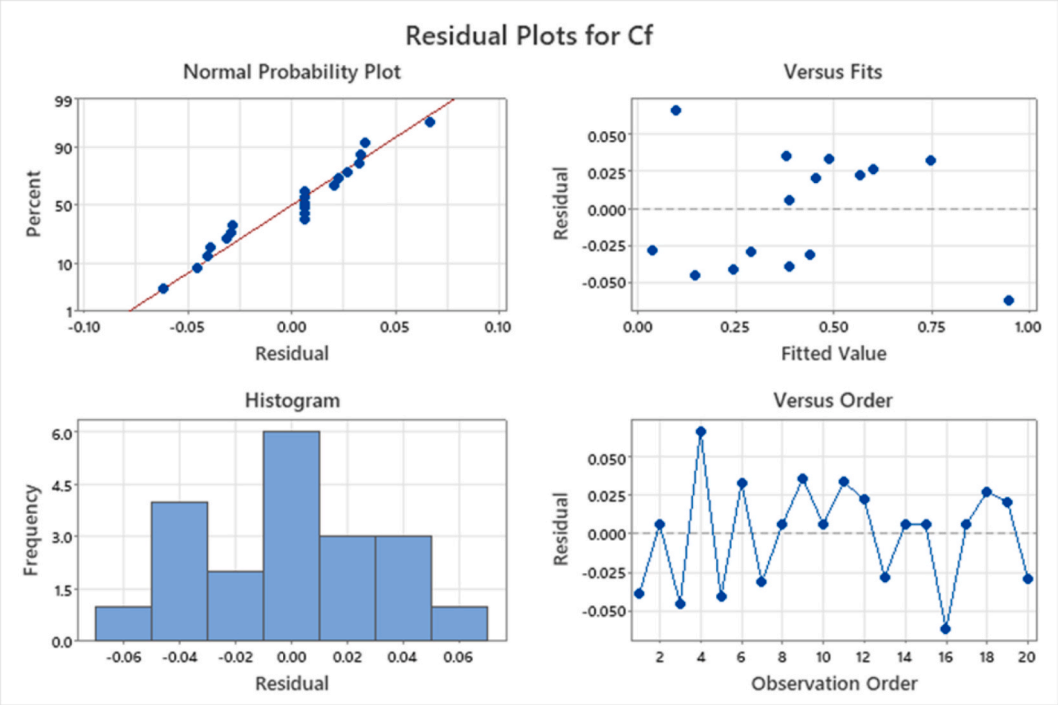


Fig. 20. Residual vs. Observation for *Cf*

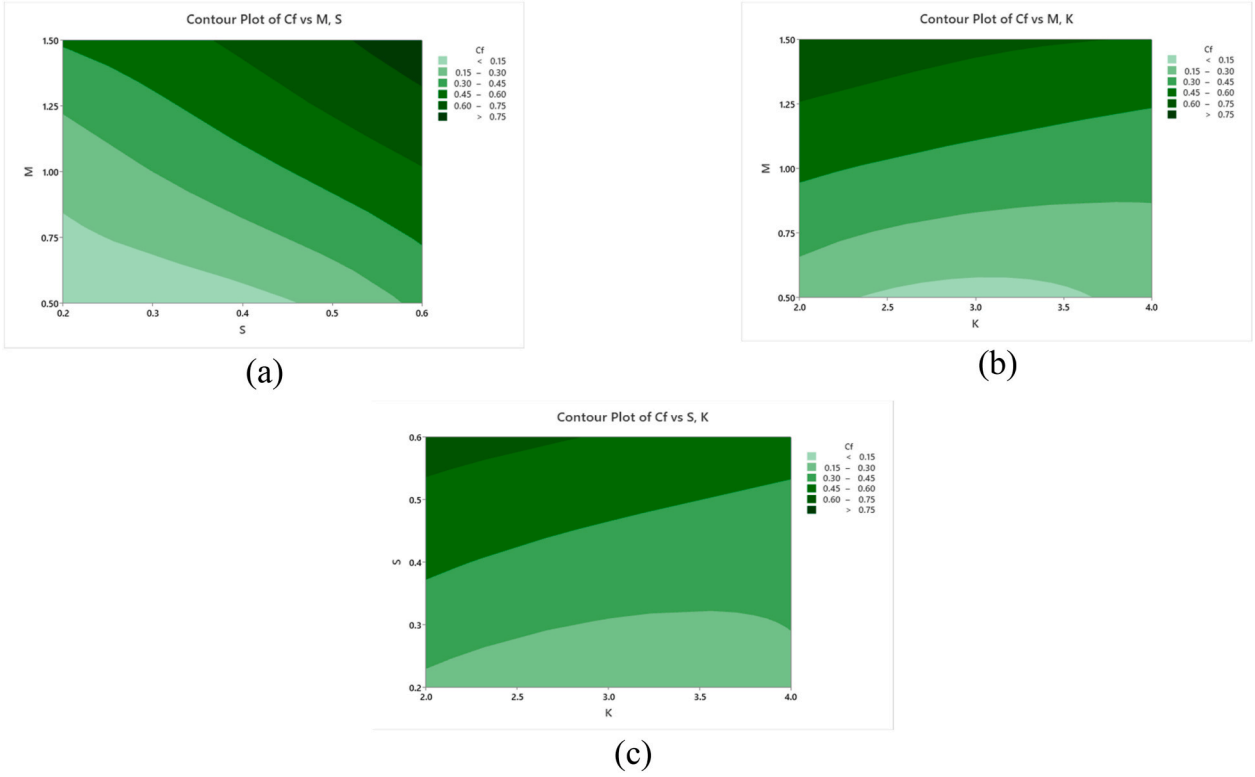


Fig. 21. Contour plots (a)-(c) depict the heat transfer characteristics of nanofluid/distilled water mixtures under various parameter interactions.

variation is observed where the heat transfer rate is high at lower R and Q_3 values, but drops sharply for increasing values. The decreasing trend in heat transfer suggests that beyond a certain threshold, the thermal boundary layer thickens, reducing the rate of conductive and convective heat exchange. This phenomenon can be attributed to the competing effects of buoyancy forces and fluid viscosity, which alter the heat dissipation mechanism. Fig. 19(c) indicates the Sherwood number $Sh(K_r, S_r)$ which represents the rate of mass transfer, exhibits a declining trend with increasing values of K_r, and S_r. Initially, the mass transfer rate is relatively high, but as these parameters increase, a downward slope is observed. The negative values in the Sherwood number suggest mass diffusion effects influenced by external conditions such as chemical

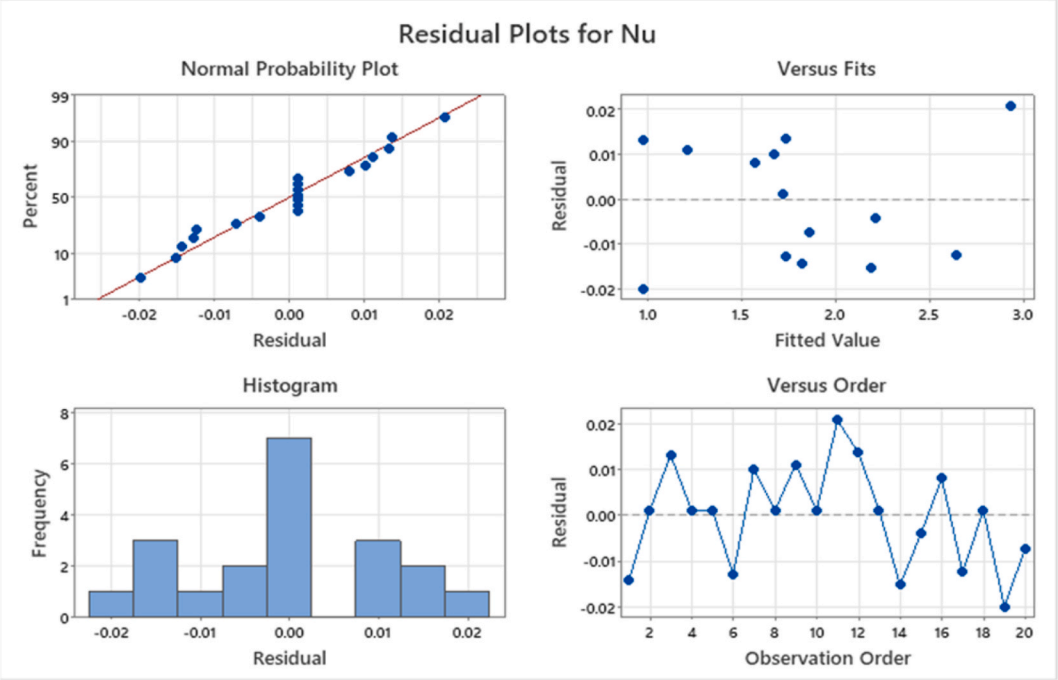


Fig. 22. Residual vs. observation for *Nu*.

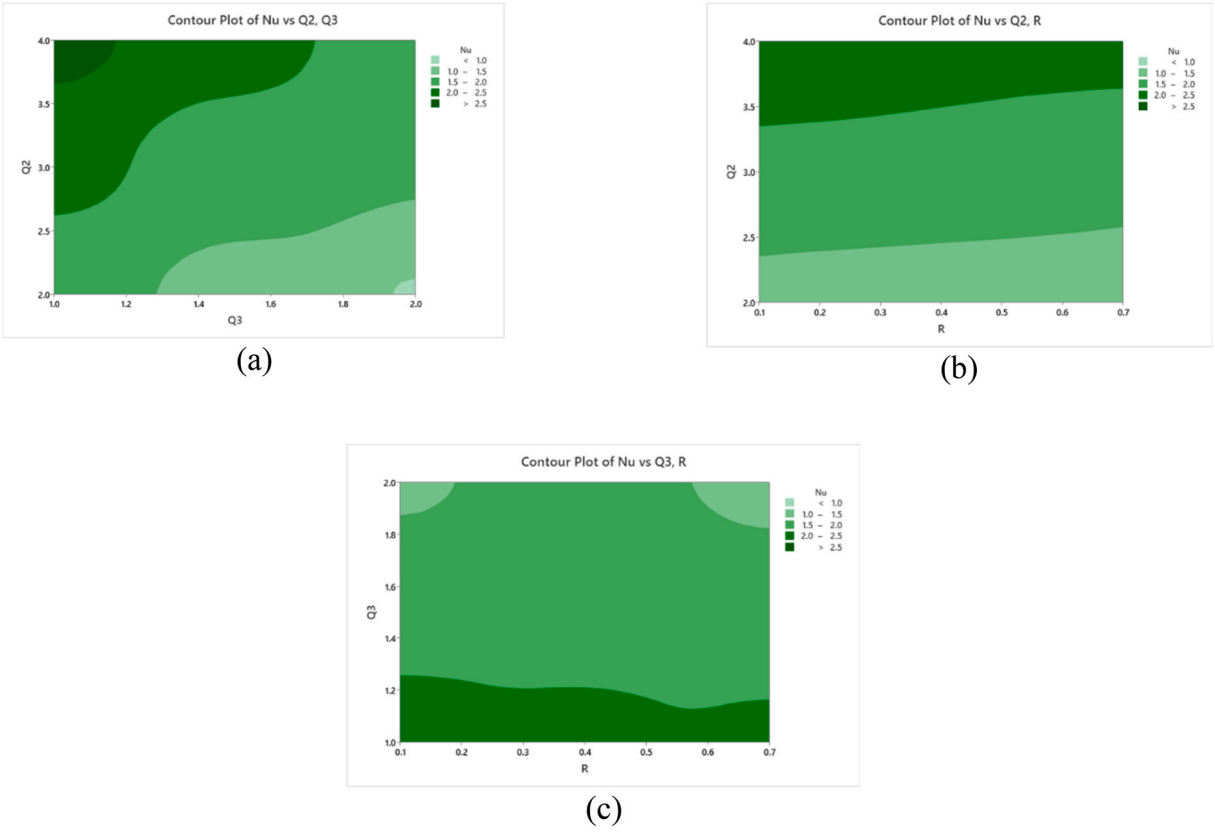


Fig. 23. Contour plots (a)-(c) depict the heat transfer characteristics of nanofluid/distilled water mixtures under various parameter interactions.

reaction rates or concentration gradients. This decline may be due to a decrease in mass flux caused by increased resistance in the concentration boundary layer (see Fig. 20-25).

To validate the accuracy of the present numerical model, a comparative analysis was conducted with the results reported by Choudhury and Ahmed (Choudhury et al., 2018). Table 7 demonstrates excellent agreement between the present work and the reference data for various values of Q_2, P_r and t, in the absence of $S = 0$, $M = 0$, $R = 0$, $Kr = 0$, $Q_3 = 0$,

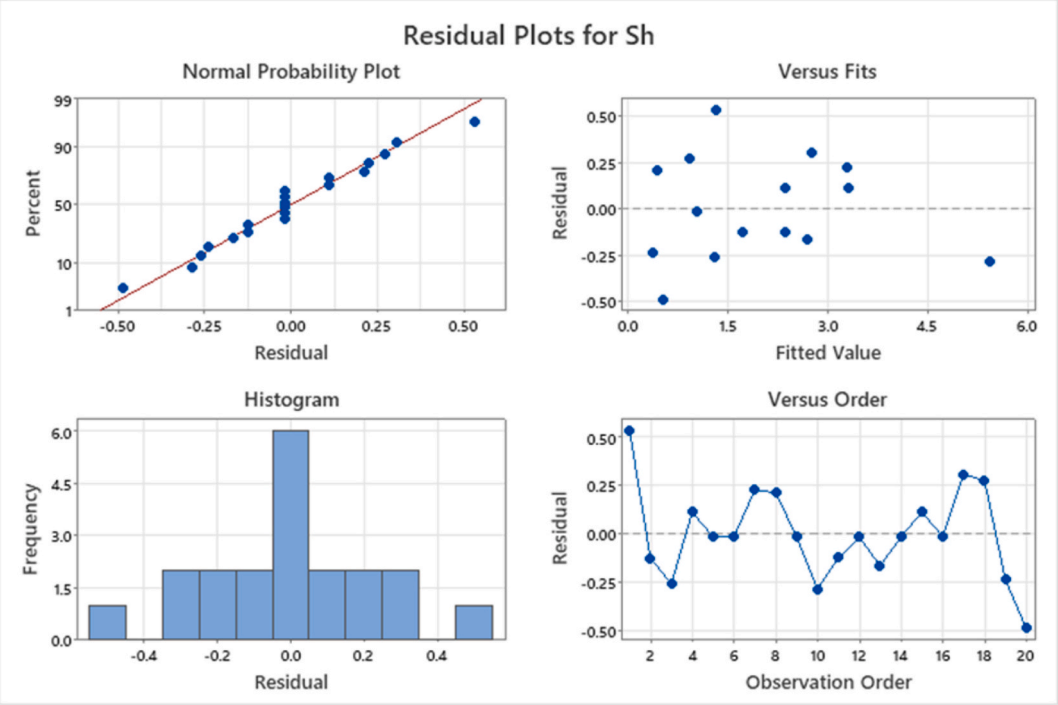


Fig. 24. Residual vs. observation for *Sh*.

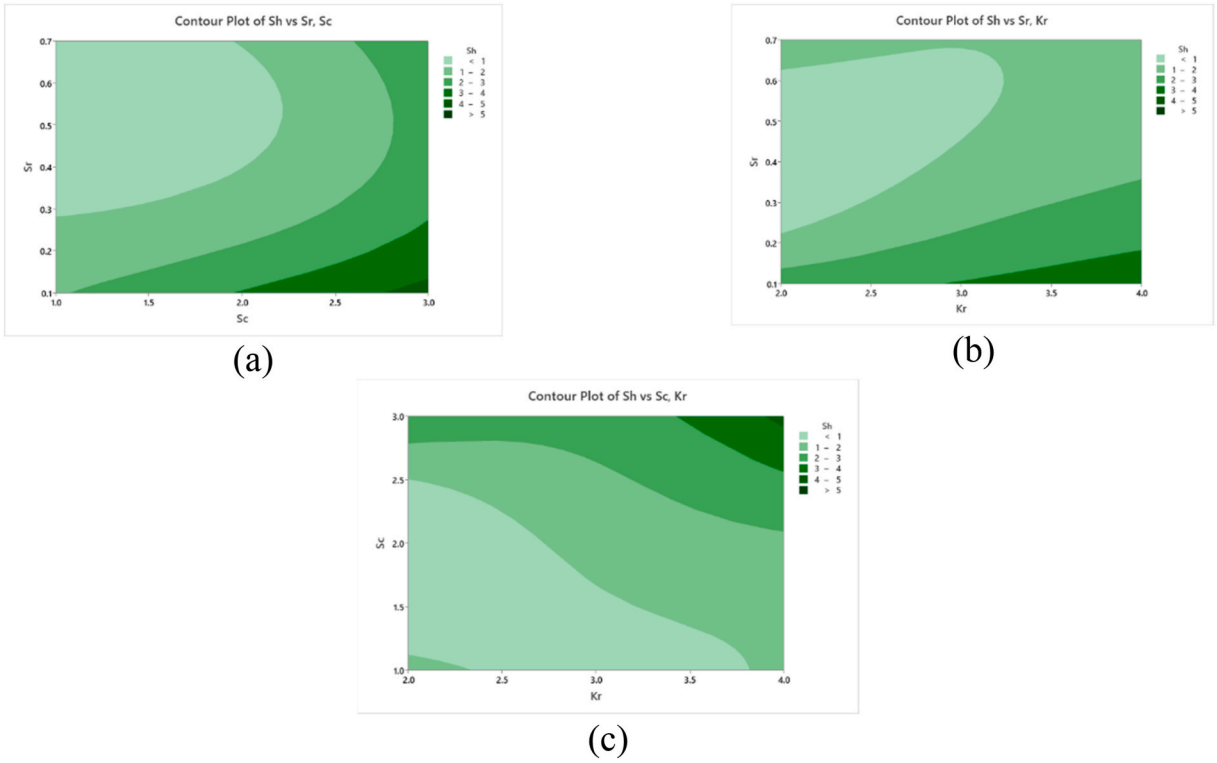


Fig. 25. Contour plots (a)-(c) depict the heat transfer characteristics of nanofluid/distilled water mixtures under various parameter interactions.

5. Response surface methodology

The RSM is a valuable tool for identifying empirical correlations between multiple variables. It is particularly useful for studying complex processes with numerous input factors, enabling the examination of their individual and interactive effects. In this study, RSM is employed to investigate the influence of key input parameters, namely M, K, and S on skin friction, Q_2, Q_3, and R on the Nusselt number, and K_r, S_r, and S_c on the Sherwood number. The input parameters were varied within specified ranges and at three levels, as presented in Tables 8, 11 and 14. Tables 9, 12 and 15 demonstrate that, following the selection of input parameters, a statistical model was developed and executed 20 times. A

Table 7
Comparison table of Nusselt number for different values of Q_2, P_r and t

Q_2	P_r	t	$-\theta'$	
			Choudhury and Ahmed. (Choudhury et al., 2018)	Present result
1	0.71	0.5	1.5791	1.5787
2	0.71	0.5	1.9695	1.9691
3	0.71	0.5	2.2778	2.2778
1	2	0.5	3.4816	3.4817
1	7	0.5	10.2180	10.2153
1	0.71	0.1	1.5665	1.5668

Table 8
Impact of varying parameter values.

Parameter	Symbol	Level		
		−1	0	1
$0.5 \leq \mathbf{M} \leq 1.5$	J_1	0.5	1	1.5
$0.2 \leq \mathbf{S} \leq 0.6$	J_2	0.2	0.4	0.6
$2 \leq \mathbf{K} \leq 4$	J_3	2	3	4

Table 9
An experimental approach utilizing CCD.

Runs	Coded values			Real values			Response
	J_1	J_2	J_3	M	S	K	Cf
1	−1	−1	−1	1.0	0.4	4	0.34709
2	1	−1	−1	1.0	0.4	3	0.39189
3	−1	1	−1	0.5	0.4	3	0.10024
4	1	1	−1	0.5	0.2	4	−0.1644
5	−1	−1	1	1.0	0.2	3	0.20207
6	1	−1	1	1.5	0.6	4	0.78204
7	−1	1	1	1.5	0.2	4	0.40962
8	1	1	1	1.0	0.4	3	0.39189
9	−1	0	0	0.5	0.6	2	0.41522
10	1	0	0	1.0	0.4	3	0.39189
11	0	−1	0	1.5	0.2	2	0.52345
12	0	1	0	1.0	0.6	3	0.59048
13	0	0	−1	0.5	0.2	2	0.00950
14	0	0	1	1.0	0.4	3	0.39189
15	0	0	0	1.0	0.4	3	0.39189
16	0	0	0	1.5	0.6	2	0.88807
17	0	0	0	1.0	0.4	3	0.39189
18	0	0	0	1.5	0.4	3	0.62943
19	0	0	0	1.0	0.4	2	0.47717
20	0	0	0	0.5	0.6	4	0.25964

Table 10
Analysis of variance.

Source	DF	SS(Adj)	MS(Adj))	F-Value	P-Value	Coefficient
Model	9	0.831938	0.092438	43.08	0.000	0.3858
Linear	3	0.798276	0.266092	124.01	0.000	-
$\boldsymbol{M}$	1	0.521474	0.521474	243.03	0.000	0.2284
$\boldsymbol{S}$	1	0.264511	0.264511	123.27	0.000	0.1626
$\boldsymbol{K}$	1	0.012291	0.012291	5.73	0.038	−0.0351
Square	3	0.009233	0.003078	1.43	0.290	-
$\boldsymbol{M}^*\boldsymbol{M}$	1	0.000381	0.000381	0.18	0.682	−0.0118
$\boldsymbol{S}^*\boldsymbol{S}$	1	0.001064	0.001064	0.50	0.497	0.0197
$\boldsymbol{K}^*\boldsymbol{K}$	1	0.003471	0.003471	1.62	0.232	0.0355
2-Way Interaction	3	0.024429	0.008143	3.80	0.047	
$\boldsymbol{M}^*\boldsymbol{S}$	1	0.006969	0.006969	3.25	0.102	0.0295
$\boldsymbol{M}^*\boldsymbol{K}$	1	0.006007	0.006007	2.80	0.125	−0.0274
$\boldsymbol{S}^*\boldsymbol{K}$	1	0.011454	0.011454	5.34	0.043	−0.0378
Error	10	0.021457	0.002146			-
Lack-of-Fit	5	0.021457	0.004291	*	*	-
Pure Error	5	0.000000	0.000000			-
Total	19	0.853396				-

Table 11
Impact of varying parameter values.

Parameter	Symbol	Level		
		−1	0	1
$2 \leq \mathbf{Q_2} \leq 4$	J_1	2	3	4
$1 \leq \mathbf{Q_3} \leq 2$	J_2	1	1.5	2
$0.1 \leq \boldsymbol{R} \leq 0.7$	J_3	0.1	0.4	0.7

Table 12
An experimental approach utilizing CCD.

Runs	Coded values			Real values			Response
	J_1	J_2	J_3	Q_2	Q_3	R	Nu
1	−1	−1	−1	3	1.5	0.1	1.8073
2	1	−1	−1	3	1.5	0.4	1.7229
3	−1	1	−1	2	2.0	0.1	0.98976
4	1	1	−1	3	1.5	0.4	1.7229
5	−1	−1	1	3	1.5	0.4	1.7229
6	1	−1	1	2	1.0	0.1	1.7283
7	−1	1	1	3	1.5	0.7	1.6858
8	1	1	1	3	1.5	0.4	1.7229
9	−1	0	0	2	1.5	0.4	1.2235
10	1	0	0	3	1.5	0.4	1.7229
11	0	−1	0	4	1.0	0.1	2.9586
12	0	1	0	4	2.0	0.7	1.7503
13	0	0	−1	3	1.5	0.4	1.7229
14	0	0	1	4	1.5	0.4	2.1764
15	0	0	0	3	1.0	0.4	2.2141
16	0	0	0	2	1.0	0.7	1.5818
17	0	0	0	4	1.0	0.7	2.6352
18	0	0	0	3	1.5	0.4	1.7229
19	0	0	0	2	2.0	0.7	0.95515
20	0	0	0	4	2.0	0.1	1.8540

Table 13
Analysis of variance.

Source	DF	SS(Adj)	MS (Adj)	F-value	P-value	Coefficient
Model	9	4.15902	0.46211	2006.78	0.000	1.72177
Linear	3	3.91880	1.30627	5672.62	0.000	-
$\boldsymbol{Q_2}$	1	2.39707	2.39707	10409.57	0.000	0.48960
$\boldsymbol{Q_3}$	1	1.46848	1.46848	6377.06	0.000	−0.41884
$\boldsymbol{R}$	1	0.05325	0.05325	231.23	0.000	−0.07297
Square	3	0.03564	0.01188	51.59	0.000	-
$\boldsymbol{Q_2}^*\boldsymbol{Q_2}$	1	0.00093	0.00093	4.03	0.072	−0.01984
$\boldsymbol{Q_3}^*\boldsymbol{Q_3}$	1	0.01066	0.01066	46.31	0.000	0.0775
$\boldsymbol{R}^*\boldsymbol{R}$	1	0.00169	0.00169	7.33	0.022	0.02676
2-Way Interaction	3	0.07003	0.02334	101.37	0.000	-
$\boldsymbol{Q_2}^*\boldsymbol{Q_3}$	1	0.04872	0.04872	211.57	0.000	−0.07804
$\boldsymbol{Q_2}^*\boldsymbol{R}$	1	0.00756	0.00756	32.85	0.000	−0.03075
$\boldsymbol{Q_3}^*\boldsymbol{R}$	1	0.01374	0.01374	59.68	0.000	0.04145
Error	10	0.00230	0.00023			-
Lack-of-Fit	4	0.00230	0.00058	*	*	-
Pure Error	6	0.00000	0.00000			-
Total	19	4.16132				-

Table 14
Impact of varying parameter values.

Parameter	Symbol	Level		
		−1	0	1
$2 \leq \boldsymbol{S_r} \leq 4$	J_1	2	3	4
$1 \leq \boldsymbol{S_c} \leq 2$	J_2	1	1.5	2
$0.1 \leq \boldsymbol{K_r} \leq 0.7$	J_3	0.1	0.4	0.7

second-order polynomial was used to illustrate the relationship between the response variables and input parameters. The model includes intercept, linear, quadratic, and interaction terms, denoted by $Z_0, Z_i,$

Table 15
An experimental approach utilizing CCD.

Runs	Coded values			Real values			Response
	J_1	J_2	J_3	S_r	S_c	K_r	Sh
1	−1	−1	−1	0.4	2	4	1.8498
2	1	−1	−1	0.1	1	4	2.2401
3	−1	1	−1	0.7	2	3	−1.0502
4	1	1	−1	0.7	3	4	3.4142
5	−1	−1	1	0.4	2	3	1.01
6	1	−1	1	0.4	2	3	1.01
7	−1	1	1	0.1	3	2	3.5145
8	1	1	1	0.4	1	3	0.65624
9	−1	0	0	0.4	2	3	1.01
10	1	0	0	0.1	3	4	5.1341
11	0	−1	0	0.1	1	2	1.6052
12	0	1	0	0.4	2	3	1.01
13	0	0	−1	0.4	3	3	2.5207
14	0	0	1	0.4	2	3	1.01
15	0	0	0	0.7	3	2	−2.4738
16	0	0	0	0.4	2	3	1.01
17	0	0	0	0.1	2	3	3.0702
18	0	0	0	0.7	1	2	−1.2054
19	0	0	0	0.7	1	4	−0.1437
20	0	0	0	0.4	2	2	0.04278

Table 16
Analysis of variance.

Source	DF	SS(Adj)	MS (Adj)	F-Value	P-Value	Coefficient
Model	9	30.1584	3.3509	31.27	0.000	1.026
Linear	3	19.4064	6.4688	60.37	0.000	-
S_r	1	5.2951	5.2951	49.42	0.000	−0.728
S_c	1	12.5588	12.5588	117.21	0.000	1.121
K_r	1	1.5526	1.5526	14.49	0.003	0.394
Square	3	8.9226	2.9742	27.76	0.000	-
S_r*S_r	1	2.8093	2.8093	26.22	0.000	1.011
S_C*S_C	1	0.7989	0.7989	7.46	0.021	0.539
K_r*K_r	1	0.0293	0.0293	0.27	0.613	−0.103
2-Way Interaction	3	1.8293	0.6098	5.69	0.015	-
S_r*S_c	1	0.0087	0.0087	0.08	0.781	−0.033
S_r*K_r	1	0.7055	0.7055	6.58	0.028	−0.297
S_c*K_r	1	1.1151	1.1151	10.41	0.009	0.373
Error	10	1.0715	0.1071			-
Lack-of-Fit	5	1.0715	0.2143	*	*	-
Pure Error	5	0.0000	0.0000			-
Total	19	31.2298				-

Z_{ii} and Z_{ij} respectively. The regression coefficients, which represent the interactions between input parameters, significantly impact the response variables. The polynomial model was fitted using Central Composite Design (CCD), which enabled 20 experimental runs, comprising six axial points, eight factorial points, and six center points. This design layout facilitated efficient estimation of regression coefficients and exploration of the factor space (Bezerra et al., 2008; Manjunatha et al., 2025).

$$Cf = Z_0 + Z_1M + Z_2S + Z_3K + Z_{11}M^2 + Z_{22}S^2 + Z_{33}K^2 + Z_{12}M^*S + Z_{13}M^*K + Z_{14}S^*K$$

To validate the proposed model and assess its practical implications, an analysis of variance (ANOVA) was performed, with the results presented in Table 10. A parameter was considered significant if its p-value was less than 0.05; otherwise, it was deemed non-significant. The high R^2 value of 97.49% indicates excellent model accuracy.

Upon examining the data, a significant relationship was found between the input variables (M, K, and S) and the skin friction coefficient (Cf), as described by the following equation: $Cf = 0.3858 + 0.2284M + 0.1626S - 0.0351K - 0.0378S^*K$.

$$Nu = Z_0 + Z_1Q_2 + Z_2Q_3 + Z_3R + Z_{11}{Q_2}^2 + Z_{22}{Q_3}^2 + Z_{33}R^2 + Z_{12}Q_2Q_3 + Z_{13}Q_2R + Z_{14}Q_3$$

To validate the proposed model and assess its practical implications, an analysis of variance (ANOVA) was performed, with the results presented in Table 13. A parameter was considered significant if its p-value was less than 0.05; otherwise, it was deemed non-significant. There is no significance for Q_2*Q_2. The high R^2 value of 99.94% indicates excellent model accuracy. Upon examining the data, a significant relationship was found between the input variables (Q_2, Q_3, and R) and the skin friction coefficient (Nu), as described by the following equation:

$$Nu = 1.72177 + 0.48960Q_2 - 0.41884Q_3 - 0.07297R + 0.0775{Q_3}^2 + 0.02676R^2 - 0.07804Q_2{}^*Q_3 - 0.03075Q_2{}^*R + 0.04145Q_3{}^*R$$

$$Sh = Z_0 + Z_1S_r + Z_2S_c + Z_3K_r + Z_{11}{S_r}^2 + Z_{22}{S_c}^2 + Z_{33}{K_r}^2 + Z_{12}S_rS_c + Z_{13}S_rK_r + Z_{14}S_cK_r$$

To validate the proposed model and assess its practical implications, an analysis of variance (ANOVA) was performed, with the results presented in Table 16. A parameter was considered significant if its p-value was less than 0.05; otherwise, it was deemed non-significant. There is no significance for K_r*K_r and S_r*S_c . The high R^2 value of 96.57% indicates excellent model accuracy. Upon examining the data, a significant relationship was found between the input variables (S_r, S_c, and K_r) and the skin friction coefficient (Sh), as described by the following equation:

$$Sh = 1.026 - 0.728S_r + 1.121\,S_c + 0.394K_r + 1.011S_r{}^*S_r + 0.539S_C{}^*S_C - 0.297S_r{}^*K_r + 0.373S_c{}^*K_r$$

6. Conclusions

This study presents a theoretical investigation of the effects of metallic nanoparticles on unsteady MHD free convective Darcy-Brinkman ternary nanofluid flow over a semi-infinite porous plate. The current study has led to the following conclusions:

- Velocity of the fluid decreases with increasing Magnetic field and suction parameter whereas permeability parameter, injection parameter and Grashof number enhances the flow.
- Radiation and temperature-dependent enhances the temperature, whereas suction and temperature gradient heat sources reduces the temperature. Nano particles volume fractions decreases the heat transfer rate from 0.15% to 0.73%, while rate of mass transfer rises from 0.53% to 1.94%.
- Higher Soret parameter enhances species diffusion, chemical reactions modify concentration by consumption, and a higher Schmidt number slows diffusion by reducing mass transfer rates. Suction decreases, while injection increases the concentration boundary layer thickness.
- RSM model validated by ANOVA with R^2 equal to 95.22%, 99.94% and 96.57% values for skin friction, heat, and mass transfer, respectively, and a significant effect of magnetic field, permeability, and Soret number.

6.1. Future work

The study suggests several avenues for future research, building upon the current investigation of unsteady Darcy-Brinkman flow of ternary nanofluids: Further research could explore the influence velocity distribution under the combined effects of Hall current and ion slip. Extending the current model to include non-Newtonian rheological behaviours (e.g., Casson (Farooq et al., 2025c), Carreau–Yasuda, or Williamson models) would provide a more realistic analysis for industrial

fluids. An entropy generation minimization approach could provide deeper insight into system irreversibility and thermal efficiency. Further extend to Entropy Generation Minimization Approach also.

CRediT authorship contribution statement

Raju B T: Writing – original draft, Visualization, Validation. **Suhana Nijamuddeen Mujawar:** Visualization, Validation, Software. **Khalil Ur Rehman:** Investigation, Formal analysis, Data curation. **Wasfi Shatanawi:** Supervision, Investigation, Formal analysis. **S.V.K. Varma:** Supervision, Software, Methodology.

Data availability statement

The data will be available upon request.

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Nomenclature

u', v'	Velocity along the x and y axis (ms^{-1})	P_r	Prandtl number
u, v	Dimensionless velocities (ms^{-1})	K_r	Chemical reaction parameter
B_0	Uniform magnetic field of strength (Am^{-1})	S_r	Soret number
$-V_0$	Normal velocity at the plate (ms^{-1})	K_T	Mean fluid temperature (K)
T'	Fluid temperature (K)	D_m	Thermal diffusion rate (m^2s^{-1})
C'	Fluid concentration $(molm^{-3})$	T_m	Mean fluid temperature
T'_∞	Free stream temperature (K)	μ_{nf}	Dynamic viscosity of nanofluid
C'_∞	Free stream concentration $(molm^{-3})$	μ_{bf}	Base fluid viscosity
T'_w	Temperature at the wall (K)	σ_{nf}	Electrical conduction of nanofluid (Sm^{-1})
S	Suction/injection parameter	σ_{bf}	Electrical conduction of base fluid (Sm^{-1})
θ	Dimensionless temperature (K)	ρ_{nf}	Nanofluid density (kgm^{-3})
ψ	Dimensionless concentration $(molm^{-3})$	ρ_{bf}	Base fluid density (kgm^{-3})
C'_∞	Concentration at the wall $(molm^{-3})$	K_{nf}	Nanofluid heat thermal conduction
g	Acceleration o account of gravity (ms^{-2})	K_{bf}	Base fluid heat thermal conduction
K'	Permeability parameter	t'	Time (s)
M	Magnetic field parameter (Am^{-1})	t	Dimensionless time (s)
S_c	Schmidt number	G_r	Thermal Grashof number
R	Radiation parameter	Nu	Nusselt number
τ	Skin-friction (Nm^{-2})	Sh	Sherwood number
q_r	Radiative heat flux (Wm^{-2})	q_1	Constant heat source
q_2	Temperature-dependent heat source	q_3	Temperature gradient-dependent heat source
Subscripts			
tnf	Ternary nano fluid		
hnf	Hybrid nano fluid		
nf	Nanofluid		
bf	Base fluid		

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